

Iman Gandomi

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EDUCATION

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- Master of Science in Electrical and Control Engineering** 2023 - Present
K. N. Toosi University of Technology (KNTU), Tehran, Iran GPA: 18.41/20
- Thesis: Development of an intelligent segmentation tracking system for skill assessment in cataract surgery videos
 - Supervisor: [Prof. Hamid D. Taghirad](#)
- Bachelor of Science in Electrical and Control Engineering** 2019 - 2023
K. N. Toosi University of Technology (KNTU), Tehran, Iran GPA: 17.65/20
- *Thesis*: Generating dataset and recognizing important regions of capsulorhexis eye surgery by neural networks.
 - Supervisor: [Prof. Hamid D. Taghirad](#)

INTERESTS

Computer Vision — Deep Learning — Machine Learning — Computer assisted surgery — LLM Agents

SKILLS

Programming	Python, C/C++, MATLAB, Dart (Flutter), LaTeX, SQL
AI & Computer Vision	Classification, Detection & Segmentation, Tracking, OpenCV, Generative AI & LLMs, Surgical Video Analysis
Libraries & Frameworks	PyTorch, OpenCV, Scikit-learn, NumPy, Pandas, Matplotlib, Git
Soft Skills	R&D Team Mentorship, Technical Project Lead, Cross-Disciplinary Collaboration, Technical Writing & Publishing, Public Speaking
Languages	Persian (Native), English (Fluent)

PROJECTS

Surgical Analysis Platform

MediversAi (Surgicise Software)

- An integrated surgical intelligence platform that automates proficiency assessment by distilling operative video into objective motion analytics and interactive performance-tracking visualizations for resident training.

Computer-Aided Segmentation Tracking in Surgery

Master's Thesis (Phase I) - Surgicise Software Component

- A temporally consistent segmentation framework leveraging memory-based propagation for robust instrument tracking despite surgical occlusions and rapid motion.

Cataract Surgery Skill Assessment System

Master's Thesis (Phase II) (NIMAD Grant for AI-based surgical skill assessment - No.4002766)

- An automated platform for quantifying surgical proficiency through kinematic analysis of instrument-tissue interactions, aligned with clinical standards for medical accuracy.

LLM-Based Surgical Reporting System

Surgicise Software Component

- A Retrieval-Augmented Generation (RAG) framework that translates objective engineering metrics into interpretable medical narratives and actionable feedback reports for residents.

Cataract Surgery Difficulty Detection Module

Surgicise Software Component

- A deep learning classification system that categorizes intra-operative difficulty to normalize skill assessment scores against case complexity for context-aware performance evaluation.

Surgical Phase Recognition & Prediction

Surgicise Software Component

- A temporal classification system identifying surgical phases via instrument utilization patterns to enable predictive workflow segmentation and targeted performance analysis.

Capsulorhexis Segmentation and Morphology Analysis

Bachelor's Thesis

- A segmentation pipeline for capsulorhexis edge to enable trajectory analysis and quantitative precision evaluation of the capsulotomy stage via morphological metrics.

EXPERIENCE

Computer Vision & AI Engineer

ARAS AI Lab — Farabi Eye Hospital

2022 - Present

Tehran, Iran

- Led the full-cycle development of a unified surgical intelligence platform, integrating multi-modal Computer Vision and NLP models for surgical education
- Collaborated with clinical specialists at Farabi Eye Hospital to validate AI outputs against medical standards
- Served as Technical Lead for student research teams, mentoring graduate and undergraduate members through the weekly meetings to guide project implementation

Technical Lead & Deep Learning Engineer

MediversAi

2024 - Present

- Spearheading the engineering of core computer vision modules for *Surgicise*, an AI-driven platform that automates surgical skill assessment to provide objective, quantitative feedback for residents
- Architecting high-performance deep learning pipelines for real-time video segmentation, ensuring the low-latency inference required for intra-operative decision support.
- Leading the translation of computer vision analytics into automated reporting frameworks to deliver actionable, data-driven performance insights for surgical residency programs

Student Committee Member

ICRoM Conference (International Conference on Robotics and Mechatronics)

2022 - Present

Tehran, Iran

- Directed the end-to-end publication lifecycle for conference proceedings, ensuring strict IEEE Xplore indexing compliance and academic integrity
- Coordinated the *Creativity & Robotics* student competitions and technical workshops, facilitating engagement between academic researchers and industrial professionals.

PUBLICATIONS

Ahmadi MJ, **Gandomi I**, Abdi P, Mohammadi SF, Taslimi A, Khodaparast M, Hashemi H, Tavakoli M and Taghirad HD, Cataract-LMM: Large-Scale, Multi-Source, Multi-Task Benchmark for Deep Learning in Surgical Video Analysis , arXiv preprint, 2025, doi:[10.48550/arXiv.2510.16371](https://doi.org/10.48550/arXiv.2510.16371).

Hatam SM, **Gandomi I**, Ahmadi MJ and Taghirad HD, AI-Powered Framework for Cataract Surgery Video Optimization , RSI Int. Conf. Robot. Mechatron. (ICRoM), 2024, doi:[10.1109/ICRoM64545.2024.10903624](https://doi.org/10.1109/ICRoM64545.2024.10903624).

Gandomi I, Vaziri M, Ahmadi MJ, Hadipour MR, Abdi P and Taghirad HD, A Deep Dive Into Capsulorhexis Segmentation: From Dataset Creation to SAM Fine-tuning , RSI Int. Conf. Robot. Mechatron. (ICRoM), 2023, doi:[10.1109/ICRoM60803.2023.10412370](https://doi.org/10.1109/ICRoM60803.2023.10412370).

Lafouti M, Ahmadi MJ, Allahkaram MS, **Gandomi I**, Lotfi F, Mohammadzadeh M, Abdi P and Taghirad HD, Surgical instrument tracking for capsulorhexis eye surgery based on siamese networks, RSI Int. Conf. Robot. Mechatron. (ICRoM), 2022, doi:[10.1109/ICRoM57054.2022.10025355](https://doi.org/10.1109/ICRoM57054.2022.10025355).

ACADEMIC EXPERIENCE (TEACHING ASSISTANT)

- **Linear Control** ([Prof. Hamid D. Taghirad](#)) Fall 2025
- **Robotics** ([Prof. Hamid D. Taghirad](#)) Spring 2025
- **Machine Learning** ([Dr. Mahdi Aliyari-Shoorehdeli](#)) Fall 2024
- **Modern Control** [Prof. Hamid D. Taghirad](#)) Spring 2024
- **Fundamentals of Intelligent Systems** ([Dr. Mahdi Aliyari-Shoorehdeli](#)) Fall 2023
- **Linear Control** ([Prof. Ali Khaki Sedigh](#)) Fall 2023
- **Industrial Control** ([Prof. Hamid D. Taghirad](#)) Spring 2023
- **Numerical Computation** ([Dr. Amir Hosein Nikoofard](#)) Fall 2022
- **Engineering Economics** ([Dr. Amir Hosein Nikoofard](#)) Spring 2022

HONORS

- Top 12 Finalist (Top 9%), LLM Agents Hackathon** May 2025
[Sharif University of Technology](#)
- Secured 12th place out of 132 teams for developing “Surgical Education Assistant.”
 - Engineered an LLM-powered agent to provide real-time, objective feedback and automated documentation for surgical training.
- Distinguished M.Sc. Seminar Award** Fall 2024
[K. N. Toosi University of Technology \(KNTU\)](#)
- Ranked in the **Top 5** of the Master’s cohort for technical presentation quality.
- Direct M.Sc. Admission Award** Summer 2023
[K. N. Toosi University of Technology \(KNTU\)](#)
- Awarded competitive waiver for the National Graduate Entrance Exam based on exceptional GPA.
- Top 10 Bachelor’s Thesis Project** Summer 2023
[K. N. Toosi University of Technology \(KNTU\)](#)
- Recognized for research excellence in neural networks and surgical datasets.
- Outstanding Internship Performance** Summer 2022
[K. N. Toosi University of Technology \(KNTU\)](#)
- Ranked among the **top 10** students for performance during industry placement.

REFERENCES

[Professor Hamid D. Taghirad](#) Email: taghirad@kntu.ac.ir
M.Sc. Thesis Supervisor
Director, Applied Robotics & AI Solutions (ARAS)
Faculty of Electrical Engineering, K. N. Toosi University of Technology, Tehran, Iran